Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 11

Full Marks: 20 *Deadline: 5:30 PM, 18-09-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a zip file which will contain all the .gp, and .pdf files of the task in their respective folders.
3) Each Gnuplot script file should be placed in its respective task folder (e.g., Task1, Task2).
4) Your zip file should be named as <Roll No>_<Name>_<Lab Activity No>.
5) Example: B24CS055_Shareen_10.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1) To learn the fundamentals of Gnuplot scripting for creating structured 2D plots and visualizations.
2) To practice generating multiplot layouts, line plots, and clustered histograms using real and synthetic datasets.
3) To develop skills in customizing plots with titles, labels, legends, styles, and color schemes for clarity.
4) To produce publication-ready outputs in PDF format with professional formatting and readability.

Task 1: Use gnuplot to generate a 2×2 multiplot layout that visualizes the behavior of four mathematical functions using only 2D plots. Each function must be clearly labeled and plotted within its own subplot.

Functions to Plot:
Define and plot the following functions:

$f_1(x) = sin(x) + cos(x) + 1$

$f_2(x) = x^2 + x^3$

$f_3(x) = 4 \cdot a \cdot x^2 \; where \; a = 2$

$f_4(x) = \frac{sin(x)}{e^{-x}}$

Plotting Requirements:

1. Arrange all four plots in a 2×2 grid layout using multiplot mode.
2. Plot each function using a 2D plot command.
3. Each subplot must include:
4. A descriptive title indicating the function plotted.
5. Clearly labeled axes.
6. A legend (key) where applicable.
7. Use appropriate line styles and scaling for clarity.
8. Use different color for subplot

Output Settings:

1. File format: PDF
2. Output file name: multiplot_2d.pdf
3. Terminal: pdfcairo
4. Page size: 20 cm × 14 cm
5. Font: "Times New Roman", size 14 pt

Deliverable:
Submit a gnuplot script named task1.gp that generates a single PDF file containing the four 2D plots arranged in a 2×2 multiplot layout, satisfying all the criteria above.

Task2: You are given the monthly stock price data of four different items over a 12-month period (Jan to Dec). Each item's data is stored in a separate .dat file as shown below. Your task is to write a Gnuplot script to generate a line plot comparing the stock price trends of all four items across the year.

Dataset Files (Input Format)

Each .dat file contains 2 columns:

1. Column 1: Month name (Jan, Feb, ..., Dec)
2. Column 2: Stock price in dollars

Example contents:

item1.dat

Jan 110
Feb 112
Mar 108
Apr 115
May 120
Jun 125
Jul 128
Aug 127
Sep 130
Oct 135
Nov 140
Dec 145

item2.dat

Jan 90
Feb 92
Mar 95
Apr 97
May 100
Jun 105
Jul 108
Aug 110
Sep 112
Oct 115
Nov 118

Dec 120

item3.dat
Jan 130
Feb 132
Mar 129
Apr 135
May 138
Jun 140
Jul 143
Aug 145
Sep 147
Oct 150
Nov 153
Dec 155

item4.dat
Jan 100
Feb 102
Mar 101
Apr 104
May 106
Jun 109
Jul 111
Aug 113
Sep 115
Oct 118
Nov 120
Dec 123

Requirements:

1. The output file should be named: stock_prices_12months.pdf.
2. Use the PDF Cairo terminal with:
3. Size: 20cm x 14cm
4. Font: "Times New Roman", size 12

5. Set the plot title to: "Monthly Stock Price Comparison (4 Items)"
6. Label the x-axis as: "Month"
7. Label the y-axis as: "Price ($)"
8. The y-axis range should be auto-scaled based on data
9. Plot all four items in a single line plot:

Item	Color	Linestyle	Line Width
1	Blue	1 (Solid)	2
2	Red	2 (Dashed)	2
3	Green	3 (Dotted)	2
4	Orange	4 (Dashed-Dotted)	2

10. Add a legend (key) positioned at the top-left corner
11. Set the x-axis tics as months (Jan, Feb, ..., Dec)
12. Rotate x-axis labels by -45 degrees for better readability

Expected Output:

1. A PDF file named stock_prices_12months.pdf showing a line plot with:
2. 4 distinct colored and styled lines representing each item's monthly stock prices.
3. Proper x and y axis labels, title, and rotated month labels.
4. A legend indicating the four items.
5. Clear visualization of price trends across the 12 months.